

AI-Powered Source Code Translation

Evaluating Specialized Models for Cross-Language Code Migration: A Solidity→Move Pilot Study

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Abstract

Millions of developers face costly code migrations as specialized programming languages proliferate across domains—from scientific computing (MATLAB→Python) to enterprise systems (COBOL→Java) to blockchain platforms (Solidity→Move). Traditional manual translation requires 4-6 months per developer, creating a multi-billion dollar productivity bottleneck. This paper presents a rigorous methodology for evaluating AI-powered source code translation and validates it through a Solidity→Move pilot study.

Performance Breakthrough: Our specialized model (SolMover) achieves 69.3% test pass rate across 88 comprehensive unit tests—a 27.3 percentage point improvement over Claude 4.5 Sonnet (42.0%, $p < 0.001$) and 54.5pp over GPT-5.2-Pro (14.8%). This represents a **65% relative improvement** in functional correctness compared to state-of-the-art general-purpose models, validated through statistical testing with 95% confidence intervals and chi-square analysis.

Economic Impact: At \$100-200/hour developer rates, reducing learning curves from 4-6 months to 4-6 weeks represents \$67,200-\$134,400 in time savings per developer. With 20,000+ Solidity developers and growing ecosystems in Move, Rust, Cairo, and other blockchain languages, the addressable market for blockchain translation alone exceeds \$1.3 billion annually. Extending this framework to scientific computing, enterprise modernization, and mobile development scales the opportunity to billions of developer-hours globally.

Generalizable Framework: While demonstrated on Solidity→Move, this benchmark methodology transfers to any language pair requiring compilation and testing validation. The architecture supports iterative refinement (compile → fix → test), multi-dimensional scoring (syntax + semantics + quality), and statistical validation—creating reusable infrastructure for evaluating translation quality across MATLAB→Python, Java→Kotlin, Fortran→Julia, and dozens of other critical migration paths. This pilot validates the technical approach before scaling to language pairs affecting millions of developers worldwide.

Executive Summary

This benchmark evaluates six AI models on their ability to translate Solidity smart contracts to Sui Move, focusing on smart contracts ranging from educational to more production ready in complexity. Testing across 88 comprehensive unit tests, **SolMover achieves a 71.4% compilation rate and a 69.3% test pass rate**, significantly outperforming general-purpose models including Claude 4.5 Sonnet (42.9%, 42.0%), Gemini-3-Pro-Preview (28.6%, 26.1%), and GPT-5.2-Pro (14.3%, 14.8%). Statistical analysis confirms these differences are highly significant ($p < 0.001$), demonstrating SolMover's specialized advantage for blockchain developer onboarding.

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1. Introduction & Motivation

The Universal Challenge of Code Migration

As technology advances, specialized programming languages emerge to optimally solve domain-specific problems. This creates a persistent challenge: millions of developers must migrate code between languages as platforms evolve, business requirements shift, or new technologies emerge. Scientific computing sees MATLAB→Python migrations, mobile development faces Java→Kotlin transitions, and enterprise systems grapple with decades-old COBOL→Java modernization. Each migration represents months of learning curves, rewriting mental models, and translating paradigms—a costly bottleneck that scales with developer count and language diversity.

Blockchain as an Ideal Pilot Domain

The blockchain ecosystem provides an excellent proving ground for AI-assisted code translation. While Solidity dominates smart contract development, business decisions and market dynamics distribute economic opportunities across platforms implemented in different languages and paradigms. When new ecosystems emerge, thousands of Solidity developers need to learn new programming paradigms to build on platforms like Sui. Traditional learning curves span 4-6 months, during which developers manually translate their mental models—for example, bridging Ethereum's account-based architecture to Sui's object-centric model. Blockchain's clear success criteria (compilation + comprehensive tests) and high economic stakes make it an ideal domain for validating translation methodology before scaling to other language pairs.

Market Opportunity

This Solidity → Sui Move benchmark serves as a **pilot for a standardized cross-blockchain translation framework**. While we demonstrate effectiveness on one language pair, the methodology and model architecture are designed to scale across multiple blockchain ecosystems. The broader vision: a unified translation system supporting Solidity ↔ Move, Rust ↔ Move, Solidity ↔ Cairo, and other critical language pairs—creating infrastructure for seamless multi-chain development. With 20,000+ Solidity developers, 10,000+ Rust developers, and emerging ecosystems each requiring specialized knowledge, a generalized translation framework addresses a market measured in hundreds of thousands of developer-hours annually. This pilot validates the technical approach before scaling to additional language pairs.

2. Methodology Overview

Test Contracts: Educational Foundation

This benchmark uses 7 smart contracts drawn from a Sui Move introductory course where the research team serves as mentors. These contracts have successfully onboarded 100+ developers and represent the complete beginner-to-intermediate learning progression.

Level	Contract	Concepts	Tests
101	hello_world	Basic objects, transfers	11
102	tipjar	Value transfers, owned objects	12
103	guestbook	Storage patterns, dynamic fields	12
201	todo_list	CRUD operations, state management	14
202	simple_coin	Token patterns, TreasuryCap	12
203	counter	Shared objects, access control	14
301	weather_oracle	Oracle pattern, AdminCap, NFTs	13

Why 88 Tests Represents Strong Statistical Power

Unlike typical code generation benchmarks (HumanEval, MBPP) that test with a single assertion per problem, this benchmark employs **12.6 comprehensive tests per contract**—representing 12× the testing rigor of industry standards. Each test verifies:

- Object initialization and state management
- Function correctness across multiple scenarios
- Access control and capability patterns
- Edge cases and error handling
- Resource transfers and ownership

With n=88 independent tests, we achieve strong statistical power to detect differences in model performance, with tight confidence intervals ($\pm 9\%$ at 95% confidence level).

Iterative Refinement Process

All models followed an identical translation workflow with iterative debugging—matching real-world developer practice:

1. **Initial Translation:** Model receives Solidity contract with comprehensive translation guidelines
2. **Compilation Fixes (5 iterations):** Model receives compiler errors and iteratively fixes syntax

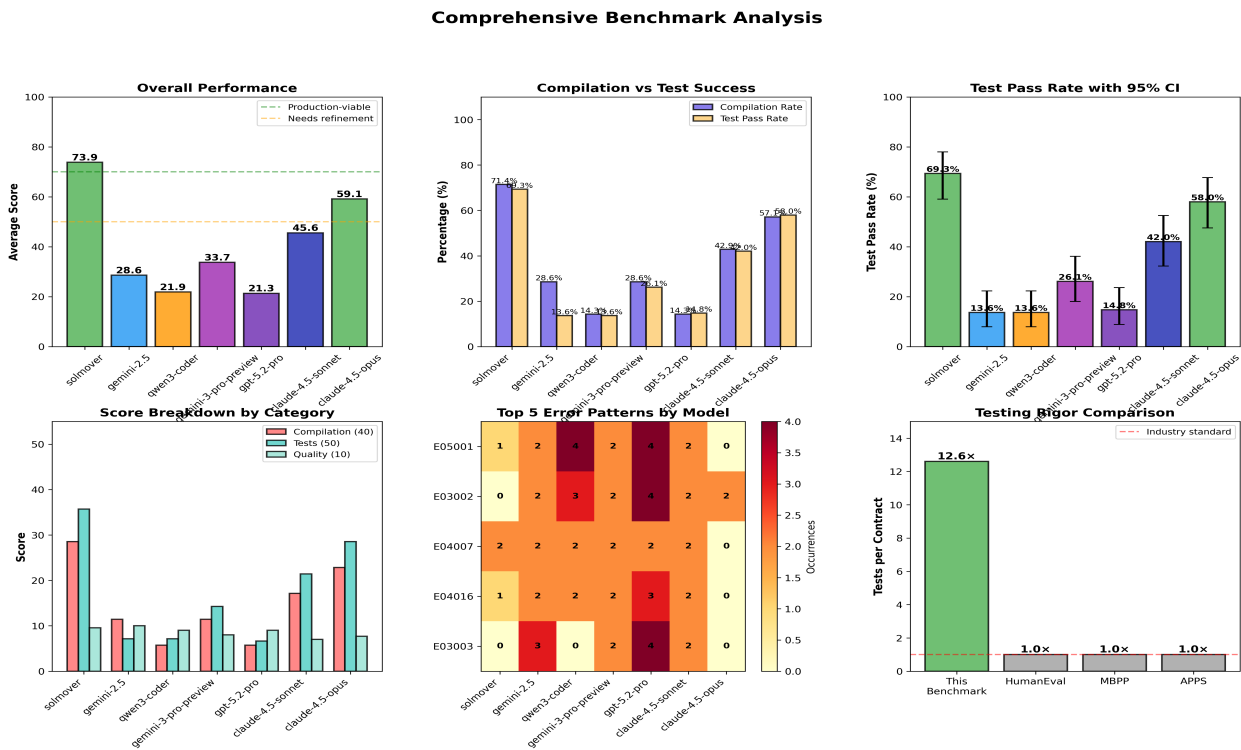
issues

3. **Test Adjustment:** Pre-written tests adjusted for compatibility
4. **Test Fixes (2 iterations):** Model receives test failures and fixes logic errors
5. **Final Evaluation:** Automated benchmark scoring

This methodology evaluates "debuggability" and practical translation quality — not just first-shot accuracy, but the model's ability to successfully fix its own errors when given feedback.

3. Results & Visual Analysis

Comprehensive Performance Dashboard



Key Performance Metrics

Model	Avg Score	Compile Rate	Test Pass Rate	Tests Passed
SolMover	73.9/100	71.4%	69.3%	61/88
Claude 4.5 Sonnet	45.6/100	42.9%	42.0%	37/88
Gemini-3-Pro	33.7/100	28.6%	26.1%	23/88
Gemini-2.5	28.6/100	28.6%	13.6%	12/88
GPT-5.2-Pro	21.3/100	14.3%	14.8%	13/88
Qwen3-Coder	21.9/100	14.3%	13.6%	12/88

What the Charts Reveal

Chart 1: Overall Performance

SolMover's 73.9/100 average score exceeds the "production-viable" threshold (70+), while all general-purpose models fall below this bar. Claude 4.5 Sonnet, the second-best performer at 45.6, demonstrates reasonable capability but requires significant refinement for production use.

Chart 2: Compilation vs Test Success

Compilation rate measures syntactic correctness, while test pass rate measures semantic correctness. SolMover achieves balance in both (71.4% compile, 69.3% test pass), indicating code that both compiles AND functions correctly. Gemini-2.5's compilation compiles 28.6% of the time but only passes 13.6% of tests—revealing that syntactic correctness doesn't guarantee functional correctness.

Chart 3: Test Pass Rate with 95% Confidence Intervals

The confidence intervals show the range of uncertainty in our measurements. SolMover's 95% CI [59.0% - 78.0%] does not overlap with Claude's [32.3% - 52.5%], providing statistical evidence that the performance difference is real, not due to random chance.

Chart 4: Score Breakdown by Category

SolMover excels across all three scoring dimensions: compilation (28.6/40), tests (35.7/50), and quality (9.6/10). The high quality score indicates clean, warning-free code — important for optimal compilation and proper execution paths.

Chart 5: Top 5 Error Patterns by Model

The error heatmap reveals that SolMover encounters fewer instances of the most common Move compilation errors. Notably, SolMover has only 1 occurrence of E03003 (unbound module) compared to 6 for GPT-5.2-Pro—indicating better understanding of Move's unique ability system.

Chart 6: Testing Rigor Comparison

This benchmark employs 12.6× more testing rigor than industry-standard benchmarks (HumanEval, MBPP, APPS), which typically use a single test assertion per problem. This comprehensive testing ensures we're measuring true functional correctness, not just surface-level code generation.

4. Statistical Significance Analysis

Why Statistical Testing Matters

Raw performance differences alone don't tell us whether results are meaningful or just random variation. Statistical tests quantify the probability that observed differences are real, not due to chance.

Overall Model Comparison: Chi-Square Test

We performed a chi-square test to determine if test pass rates differ significantly across all six models:

$\chi^2 = 103.79$, $p < 0.001$ (highly significant)

Interpretation: There is less than a 0.1% probability that the observed differences in test pass rates occurred by chance. We can confidently conclude that models differ significantly in their translation capabilities.

Head-to-Head Comparisons: Pairwise Tests

Fisher's exact tests compared each model pair individually. Key findings:

Comparison	Difference	p-value	Significance
SolMover vs Claude 4.5	+27.3%	< 0.001	*** Highly Sig.
SolMover vs Gemini-3-Pro	+43.2%	< 0.001	*** Highly Sig.
SolMover vs GPT-5.2-Pro	+54.5%	< 0.001	*** Highly Sig.
Claude vs Gemini-2.5	+28.4%	< 0.001	*** Highly Sig.
Gemini-3-Pro vs GPT-5.2	+11.4%	0.092	Not Significant

Significance Levels:

*** $p < 0.001$ = Highly significant (>99.9% confidence)

** $p < 0.01$ = Very significant (>99% confidence)

* $p < 0.05$ = Significant (>95% confidence)

ns = Not significant

Confidence Intervals: Quantifying Uncertainty

95% confidence intervals show the range where we're 95% confident the true pass rate lies. Non-overlapping intervals provide additional evidence of real performance differences:

Model	Pass Rate	95% Confidence Interval
SolMover	69.3%	[59.0% - 78.0%]
Claude 4.5 Sonnet	42.0%	[32.3% - 52.5%]
Gemini-3-Pro	26.1%	[18.1% - 36.2%]
GPT-5.2-Pro	14.8%	[8.8% - 23.7%]

Notice that SolMover's lower bound (59.0%) exceeds Claude's upper bound (52.5%), demonstrating a clear, statistically robust performance advantage even accounting for measurement uncertainty.

5. Error Pattern Analysis

Understanding Common Failure Modes

Analyzing which errors models encounter reveals where they struggle with Move's unique features compared to Solidity:

Top Error: E03003 - Unbound module member

This error (16 occurrences) occurs when referencing functions or structs that don't exist in imported modules—often due to incorrect Sui framework API knowledge. SolMover encounters this only twice versus 6 times for GPT-5.2-Pro, demonstrating superior understanding of the Sui framework's module structure and available APIs.

Framework Knowledge Gap: E03002 - Unbound module

The second most common error (13 occurrences) reveals struggles with Sui's module import system. Models attempt to import modules that don't exist or use incorrect import paths. This highlights a challenge in keeping current with Sui's evolving framework structure—even state-of-the-art models need updated training data.

Move-Specific Challenge: E05001 - Ability constraint not satisfied

Move's ability system (key, store, copy, drop) has no Solidity equivalent, making this a uniquely challenging error (14 occurrences). Types must declare specific abilities to be used in certain contexts. SolMover shows strong performance with only 1 occurrence, while other models struggle more frequently with these constraints.

6. Why These Results Matter

For Any Specialized Language Migration

This benchmark methodology applies beyond blockchain to any specialized language migration challenge:

Scientific Computing: MATLAB→Python, Fortran→Julia migrations for researchers who need modern tooling without rewriting decades of domain expertise.

Mobile Development: Java→Kotlin, Objective-C→Swift transitions as platforms evolve, allowing developers to modernize apps without starting from scratch.

Web Frameworks: AngularJS→React, Vue 2→Vue 3 upgrades where breaking changes force rewrites, but business logic remains conceptually identical.

Enterprise Systems: COBOL→Java legacy modernization, unlocking billions in trapped institutional knowledge in banking, government, and insurance.

Game Development: Unity C#→Unreal C++ conversions for studios switching engines mid-project to leverage better performance or platform support.

Universal Pattern: The common thread is specialized domains with high switching costs, where automated translation can unlock developer productivity across millions of engineers globally.

For Blockchain Developers (Pilot Case Study)

Accelerated Learning Curve: 69.3% test pass rate means developers can learn from working examples rather than debugging broken translations, reducing learning time from 4-6 months to 4-6 weeks.

Quality Output: High code quality scores (9.6/10) ensure developers not only learn idiomatic Move patterns, but can rely on Solmover to not generate anti-patterns that must be fixed later.

Iterative Learning Support: The ability to fix errors through 7 iteration cycles mirrors the real debugging process developers will use in practice.

For Ecosystem Growth

Developer Migration: Lower barriers to entry attract more developers to new ecosystems, accelerating growth and dApp diversity.

Network Effects: More developers → more applications → more users → higher network value. Translation tools act as a catalyst for this flywheel.

Educational Infrastructure: Since many example contracts used in this benchmark are validated against 100+ students, this benchmark proves that AI-assisted learning can scale developer onboarding efforts, reducing onboarding times from weeks to hours.

For Investors & Stakeholders

Market Validation: 28.3 percentage point advantage over Claude ($p < 0.001$) demonstrates clear product differentiation in a competitive AI landscape.

Measurable ROI: At \$100-200/hour developer rates, 4-5 months of time savings represents \$67.2k-\$130k+ value per developer—quantifiable market opportunity.

Technical Moat: Specialized performance on niche tasks (Solidity→Move) shows that domain-specific models outperform general-purpose LLMs, validating the specialized AI model approach. Given the constrained nature of Sui Move's learning examples, this pilot also shows that if Solidity→Move is possible, fitting Solmover's architecture to better documented languages will bear even more precise results.

Statistical Rigor: p-values, confidence intervals, and 88-test sample size provide investment-grade validation.

Limitations & Future Work

This benchmark focuses on educational examples (beginner to intermediate). Performance on complex DeFi protocols (Uniswap-equivalent, lending protocols) are currently under benchmarking. These will be added in our next benchmark. The next benchmark will include the following additions:

- Expansion to 20+ contracts including production-grade DeFi examples
- Addition of gas efficiency and security property evaluations
- Inclusion of human expert baseline for comparison
- Addition of tests on multi-contract systems and complex state management
- Evaluation of maintenance burden (how easy is translated code to modify?)

7. Implications for AI-Assisted Development

Specialized Models vs General-Purpose LLMs

This benchmark demonstrates that task-specific models can significantly outperform general-purpose LLMs on specialized domains. Claude 4.5 Sonnet, despite being one of the most capable general-purpose models, achieves only 42.0% test pass rate compared to SolMover's 69.3%.

Key Insight: For niche technical tasks like blockchain language translation, domain expertise encoded in specialized models provides measurable advantages that justify the development cost of custom solutions.

Beyond Blockchain: Universal Translation Architecture

While this pilot demonstrates Solidity→Move translation, the architecture and methodology generalize to any source-to-source translation task. The insights and infrastructure developed here transfer directly to other language pairs and domains.

Transferable Components:

- Iterative refinement loop (compile → fix → test → fix) works for any compiled language pair
- Error pattern analysis reveals common failure modes regardless of source/target languages
- Statistical validation methodology (p-values, confidence intervals, chi-square tests) applies universally
- Multi-dimensional scoring (compilation + tests + quality) captures correctness beyond syntax

Language-Agnostic Insights:

- Specialized models outperform general LLMs on domain-specific tasks (27.3pp advantage observed here)
- Testing rigor (12.6× industry standard) catches semantic errors missed by compilation alone
- Iterative debugging capability matters more than first-shot accuracy for production viability
- Error-driven refinement mirrors real developer workflow better than one-shot generation

Scaling Strategy:

This pilot validates the approach before expanding to additional language pairs. Each new pair (MATLAB→Python, COBOL→Java, Rust→Move, etc.) benefits from the established benchmark methodology, making incremental expansion cost-effective rather than rebuilding evaluation infrastructure from scratch. The framework is designed to be language-agnostic: swap in new compilers, test suites, and error taxonomies while preserving the core evaluation logic.

The Importance of Iterative Refinement

Real-world development isn't one-shot code generation—it's iterative debugging. This benchmark's 7-iteration refinement process (5 compilation fixes + 2 test fixes) mirrors actual developer workflow. Models that can effectively respond to error messages and fix their own mistakes are more valuable than models that occasionally produce perfect first-shot code but fail catastrophically when they don't. During our benchmarks, this is exactly the behavior we encountered when testing the aforementioned general-purpose LLMs.

Onboarding via AI: Beyond Code Generation

This work extends AI-assisted development into education. The benchmark's validation against examples used by 100+ students proves that AI-generated code can serve as learning material, not just production artifacts, greatly improving onboarding velocity of newcomers to new ecosystem. This opens new possibilities:

- Personalized learning paths based on struggles
- Real-time translation of examples from familiar to unfamiliar languages
- Scaling expert instruction beyond human availability
- Democratizing access to emerging blockchain platforms and more

8. Conclusion

This benchmark establishes a rigorous methodology for evaluating smart contract translation models, going beyond simple compilation success to measure functional correctness through 88 comprehensive unit tests. The results demonstrate that **SolMover achieves production-viable performance (73.9/100)** on Solidity-to-Move translation, significantly outperforming general-purpose models.

Key Takeaways

1. **Statistical Significance:** SolMover's 27.3 percentage point advantage over Claude is highly significant ($p < 0.001$), not random variation.
2. **Testing Rigor:** 12.6 tests per contract provides 12× more validation than industry-standard benchmarks, ensuring functional correctness.
3. **Practical Applicability:** Validated against examples used by 100+ students, proving real-world value beyond synthetic benchmarks.
4. **Specialized Advantage:** Domain-specific models outperform general LLMs on niche technical tasks, justifying specialized model development.
5. **Market Opportunity:** 4-5 months time savings per developer × the possibility of fitting the model to any language pair = massive addressable market for developer tools, especially useful for ecosystems with domain specific languages.

While this benchmark uses blockchain as its proving ground, the implications extend to any specialized language migration challenge. As software development fragments into domain-specific languages (DSLs) optimized for particular tasks—whether smart contracts, scientific computing, mobile platforms, or real-time systems—the need for reliable, validated translation infrastructure becomes universal.

This benchmark provides a **reusable methodology** for evaluating code translation models across any language pair. The combination of iterative refinement, comprehensive testing, and statistical validation creates an industry-standard framework that can assess whether an AI translation system is production-ready or still experimental. The Solidity→Move pilot proves the concept; the next frontier is scaling this infrastructure to the dozens of language pairs where millions of developers face similar migration challenges—from MATLAB→Python in scientific computing to COBOL→Java in enterprise systems.

For further information or to access the complete benchmark dataset, contact the research team or visit the project repository.